

28. MEDICAL DIAGNOSIS

AIM

To develop a medical diagnosis system using Prolog that identifies a possible disease based on the symptoms given by the user.

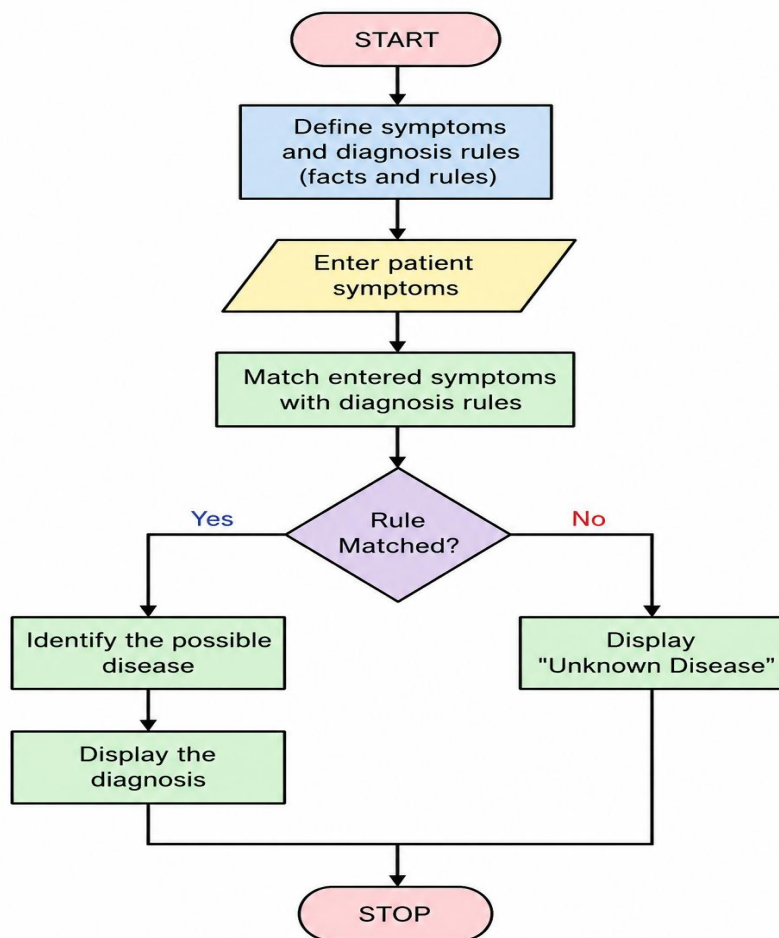
OBJECTIVE

- To represent medical knowledge using Prolog facts and rules.
- To accept symptoms as input.
- To apply logical inference for diagnosis.
- To display the possible disease based on matching symptoms.

ALGORITHM

1. Start the program.
2. Define symptoms as Prolog facts.
3. Define diagnosis rules for different diseases.
4. Enter the patient's symptoms.
5. Match the symptoms with the predefined rules.
6. If a rule matches, identify the possible disease.
7. Display the diagnosis.
8. If no rule matches, display an unknown diagnosis.
9. Stop the program.

FLOWCHART



CODE

% Medical Diagnosis System

diagnosis(flu) :-

 symptom(fever),

 symptom(cough),

 symptom(body_pain).

diagnosis(cold) :-

 symptom(cough),

 symptom(runny_nose),

 symptom(sneezing).

diagnosis(malaria) :-

```
symptom(fever),
symptom(chills),
symptom(sweating).

diagnosis(typhoid) :-
    symptom(fever),
    symptom(headache),
    symptom(abdominal_pain).

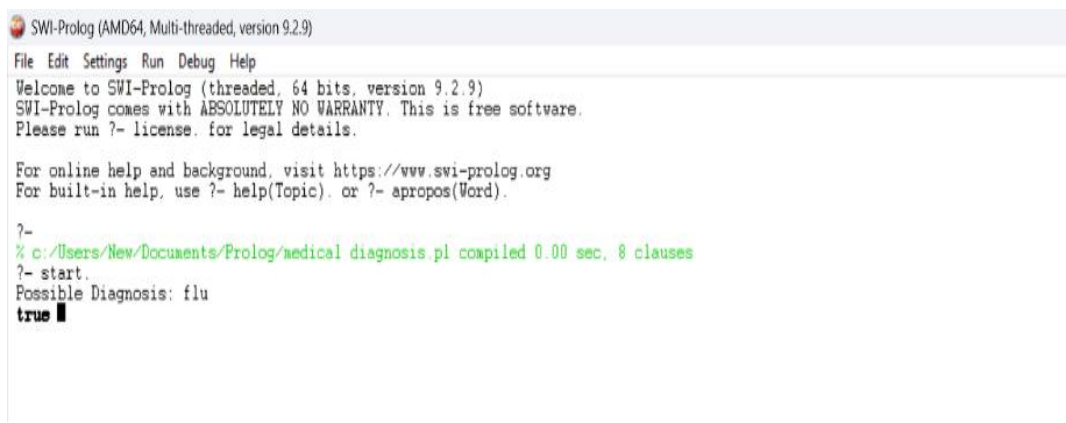
symptom(fever).

symptom(cough).

symptom(body_pain).

start :-
    diagnosis(Disease),
    write('Possible Diagnosis: '),
    write(Disease),
    nl.
```

OUTPUT



```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/medical diagnosis.pl compiled 0.00 sec, 8 clauses
?- start.
Possible Diagnosis: flu
true
```

RESULT

The Prolog program for medical diagnosis was successfully implemented. The system uses symptoms, facts, and logical rules to identify a possible disease based on the given symptoms.